

EXTRACTION

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DEFINITION

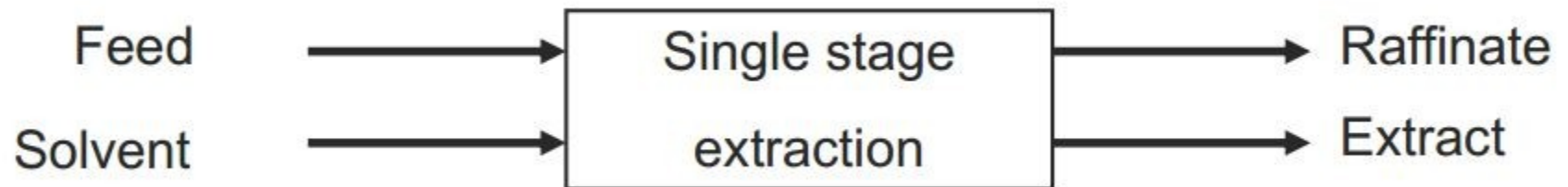
- **Extraction** – is the process by which a solute is transferred from one phase to a new phase.
- is also the process of obtaining the medicinally important components of a plant or a crude drug
- The products so obtained from plants are relatively impure liquids, semisolids or powders

OPERATING MODES OF EXTRACTION

- Mode of extraction could be
 - Batch extraction – single stage or multiple stage
 - Continuous extraction

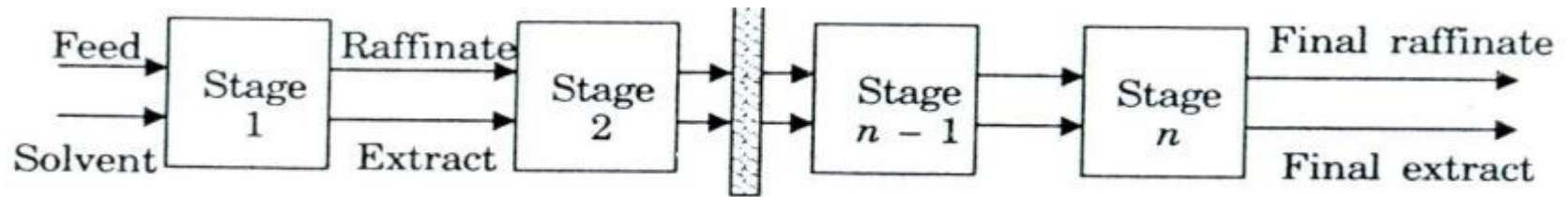
BATCH EXTRACTION

- The crude drug is mixed with solvent
- After equilibration, the desired solute is separated out for further processing.
- A schematic representation of a single batch operation:



CONTINUOUS EXTRACTION

- There is a constant flow of new solvent
- Constant interaction of crude drug and solvent
- Does not achieve equilibrium



IDEAL ASPECTS OF AN EXTRACTION METHOD

- should preserve the plant constituents
- Preferably fast
- Should be efficient
- Should be economical
- Should be selective to the medicinally important plant components

SOLVENT EXTRACTION

- is a method of extraction where the plant material is extracted with a solvent
- could be liquid, semisolid or supercritical fluid (CO₂ or N₂)
- is mainly due to solubility and to some extent due to diffusion
- the choice of solvent depends on:
 - "like dissolves like"
 - Polarity index

A VOLATILE SOLVENT EXTRACTION

- Preparation of plant material
 - Collection
 - Garbling
 - Drying
 - In most cases, the plant material is dried but some times the **fresh plant** material is extracted
 - Grinding
 - facilitates extraction by increasing surface area

DRYING

○ Aim of drying:

- Ease of transport and storage
- Ease of grinding
- Inhibit the growth of microorganisms
- Preservative of active constituents

○ Drying is done in:

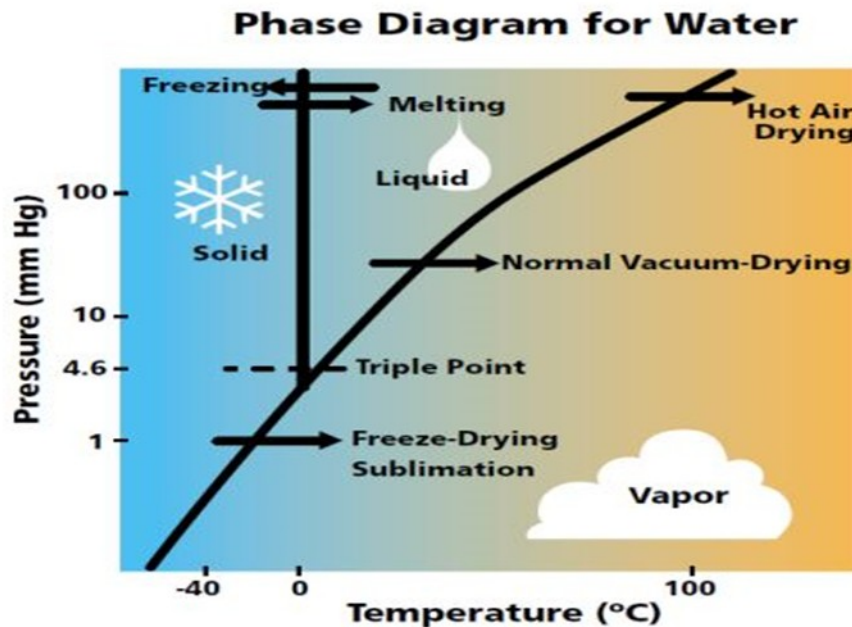
- Shade and in sunlight (Natural drying)
- Hot air drying or by freeze-drying (Artificial drying)

FREEZE DRYING (LYOPHILISATION)

- used to dry extremely heat – sensitive materials.
 - proteins, blood products
- The freeze drying process is divided into two different physical process steps:
 - Step one is the freezing of the material below its solidification temperature

LYOPHILISATION....

- Step two removes the ice crystals at very low temperatures.
 - First by sublimation then desorption



HOT AIR DRYING METHOD

- Oven-drying
 - uses thermal energy to remove moisture from the samples.
 - is one of the easiest and rapid thermal processing that can preserved phytochemicals

OVEN



CHEMICAL DRYING USING DESICCATORS

- An absolutely dried drug is that completely freed from water
 - when exposed to air it absorbs 8-10% of moisture and is called air-dry drug

DESICCATORS



MACERATION

- The whole or coarsely powdered crude drug is placed in a stoppered container with the solvent
- It is then allowed to stand at room temperature for a period of at least 3 days with frequent agitation
- Then the solution is filtered
 - Filtrate---Extract
 - Residue---Marc



MACERATION...

○ Advantage

- is the easiest and simple method
- Preserve thermolabile compounds

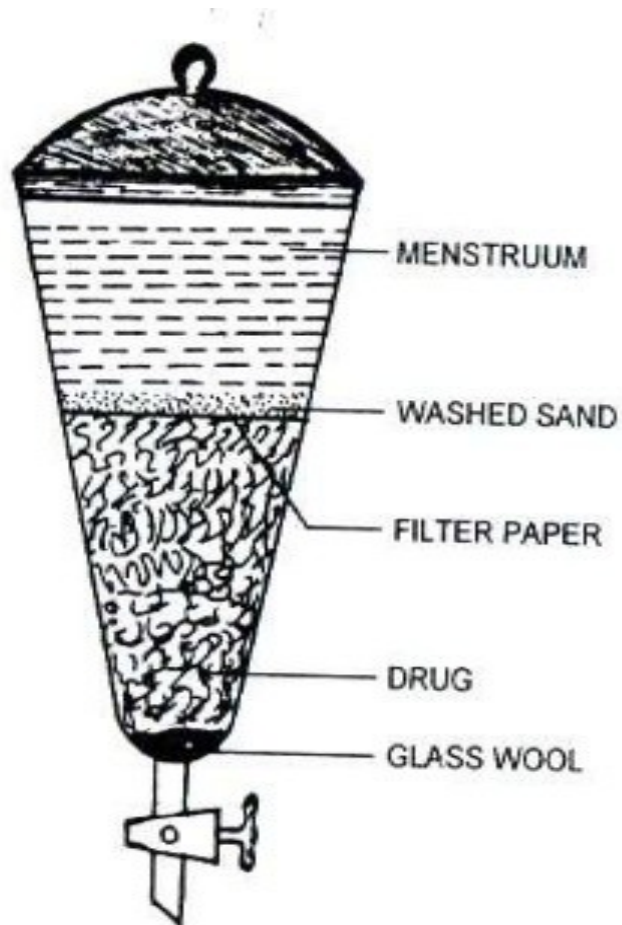
○ Disadvantage

- Consumes too much solvent
- Saturation is attained quickly.
- Continuous stirring is needed .
- Time taking process with the possibly of hydrolysis of compounds

PERCOLATION

- Slow passage of menstruum under the influence of gravity through a column of the drug powder
- The solid ingredients are moistened with an appropriate amount of the specified menstruum and allowed to stand for approximately 4 h in a well closed container
- Then crude drug is packed and the top of the percolator is closed.

PERCOLATOR



PERCOLATION

- Additional menstruum is added to form a shallow layer above the mass, and the mixture is allowed to macerate in the closed percolator for 24 h.
- The outlet of the percolator then is opened and the liquid contained therein is allowed to drip slowly.

PERCOLATION

○ Advantage

- is the easiest and simple method
- Preserve thermolabile compounds

○ Disadvantage

- Consumes too much solvent
- Continuous stirring is needed .

INFUSION

- Infusion is extraction by hot solvent and the contact time between the plant material and the solvent is relatively short.
 - Good for fragile plant parts like leaves, buds or flowers.

INFUSION



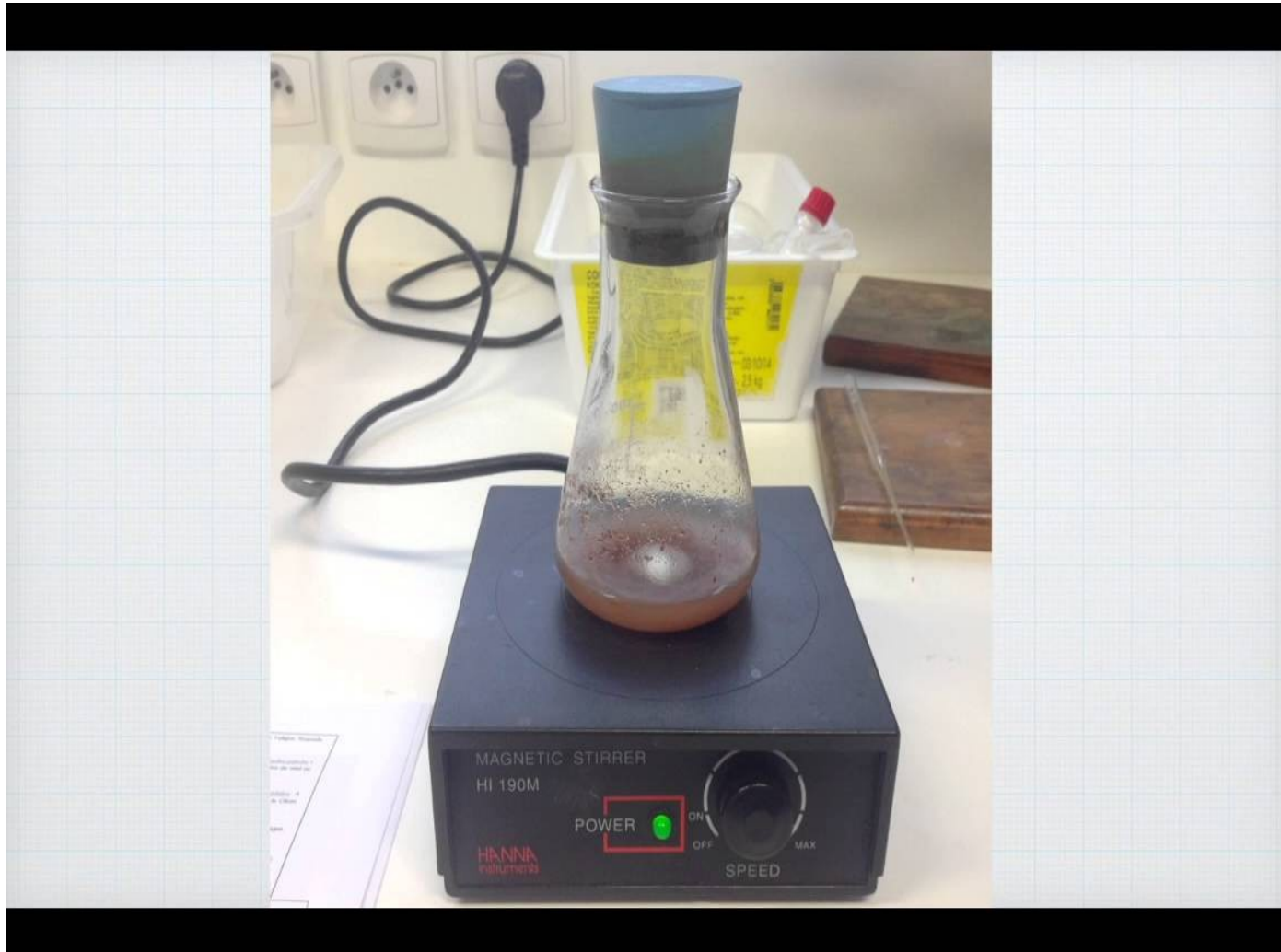
INFUSION

- Advantage:
 - Simplicity
 - Quick method
- Disadvantage:
 - Sensitive Thermolabile compounds will be destroyed
 - Wastage of solvent

DECOCTION

- Decoction is extraction by boiling. The plant material is boiled in the solvent.
 - Good for woody parts
- Advantage:
 - Simplicity
 - Quick method
- Disadvantage:
 - Thermolabile compounds will be destroyed
 - Wastage of solvent

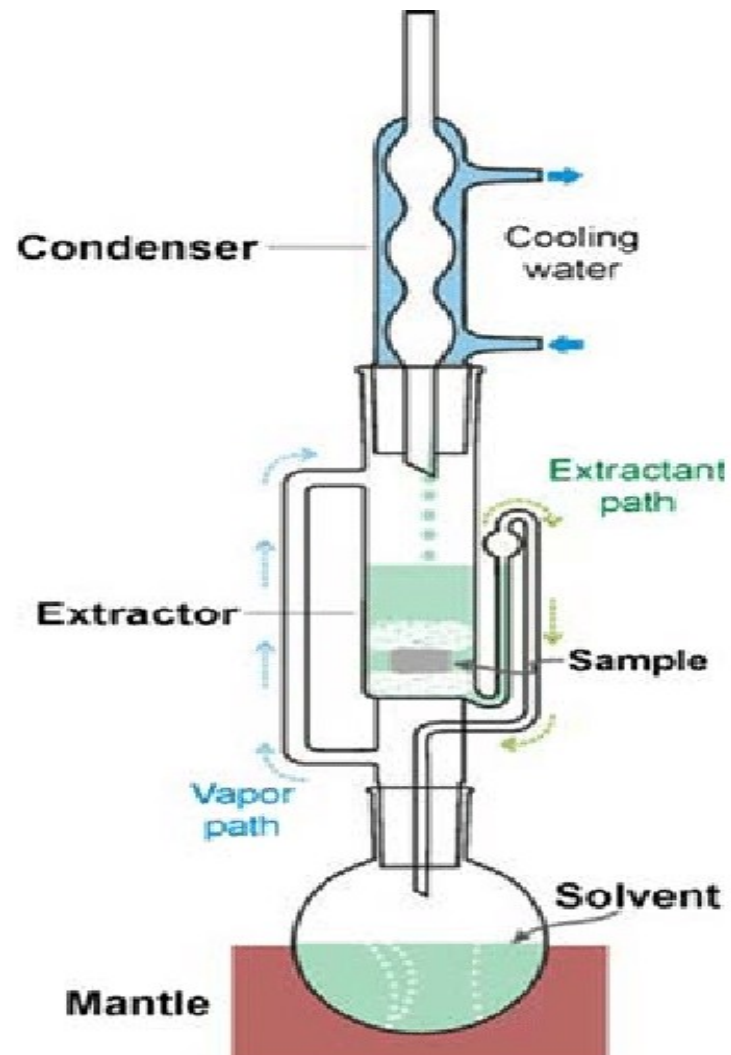
DECOCTION



SOXHLET (CONTINUOUS) EXTRACTION

- Powdered plant material is packed in the thimble and the mouth of the thimble is covered by piece cotton.
- The packed thimble is put in the extraction chamber and enough amount of volatile solvent is added to the flask.
- After all the components are fitted appropriately, water is let to flow through the condenser and the heater is switched on.

SOXHLET



SOXHLET

- Advantage:

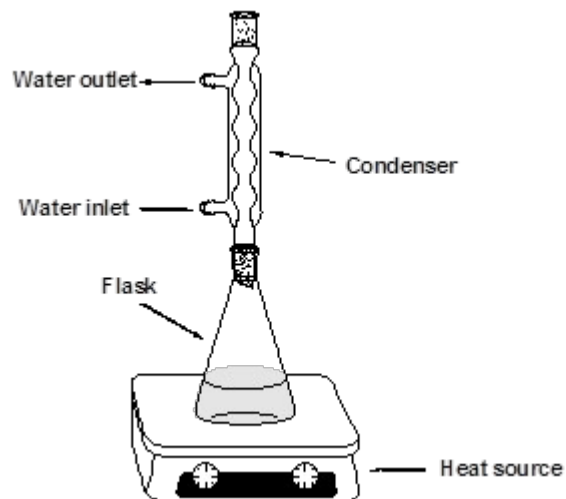
- Saves solvent
- Quick method
- Ease of changing solvent for the preparation of fractions

- Disadvantage:

- Requires special apparatus
- Not applicable for thermolabile plant constituents
- No direct contact between plant material and heat source

REFLUXING

- Similar to decoction except that there is a condenser
- It is a constant volume extraction method because there is no wastage of solvent.



REFLUXING

- Advantage:
 - Saves solvent
 - Quick method
- Disadvantage:
 - Not applicable for thermolabile plant constituents

SOLVENTS USED IN EXTRACTION

- The ideal solvent for a certain pharmacologically active constituent should
 - Be highly selective for the compound to be extracted
 - Have a high capacity for extraction in terms of coefficient of saturation of the compound in the medium
 - Not react with the extracted compound or with other compounds in the plant material
 - Have a low price
 - Be harmless to man and to the environment
 - Be completely volatile

CHOICE OF SOLVENTS...

- For intracellular plant constituents
 - 80% methanol ("Universal solvent") dissolves non-polar and polar compounds.
- For extracellular plant constituents

CHCl₃, CH₂Cl₂, CH₃COCH₃, C₂H₅OH, C₆H₁₄,
petroleum ether, hydroalcoholic solutions...
- Generally, extraction of a given plant material starts with non polar (defating)

NON-VOLATILE SOLVENT EXTRACTION

- The enfleurage method
- The pneumatic method
- The maceration method

THE ENFLEURAGE METHOD

- Several frame containing fat are placed one above the other so that the flowers get sandwiched between the two layers which absorbs the oil as it is given off.
- Exhausted flowers are removed and replaced by the fresh flowers until the fat is saturated with flower oil

ENFLEURAGE



THE PNEUMATIC METHOD

- involves the passage of a current of warm air through the flowers.
- The air laden with suspended volatile oil is then passed through melted fat in which the volatile is dissolved and absorbed.

THE MACERATION METHOD

- The flowers are gently heated in melted fat
- Rotated in the fluid until exhausted, then they are strained out, squeezed
- the exuded fluid is returned to the fat and so on until a special concentration is reached.
- The volatile oil obtained by 3 successive extractions with alcohol from the fat.

SUPER CRITICAL FLUID EXTRACTION

- Beyond its critical point, a fluid can have the density of a liquid and the viscosity of a gas,
 - Therefore it diffuses well through solids, and it is a good solvent.
- Although, in theory, several gases can be used, the focus is exclusively on carbon dioxide

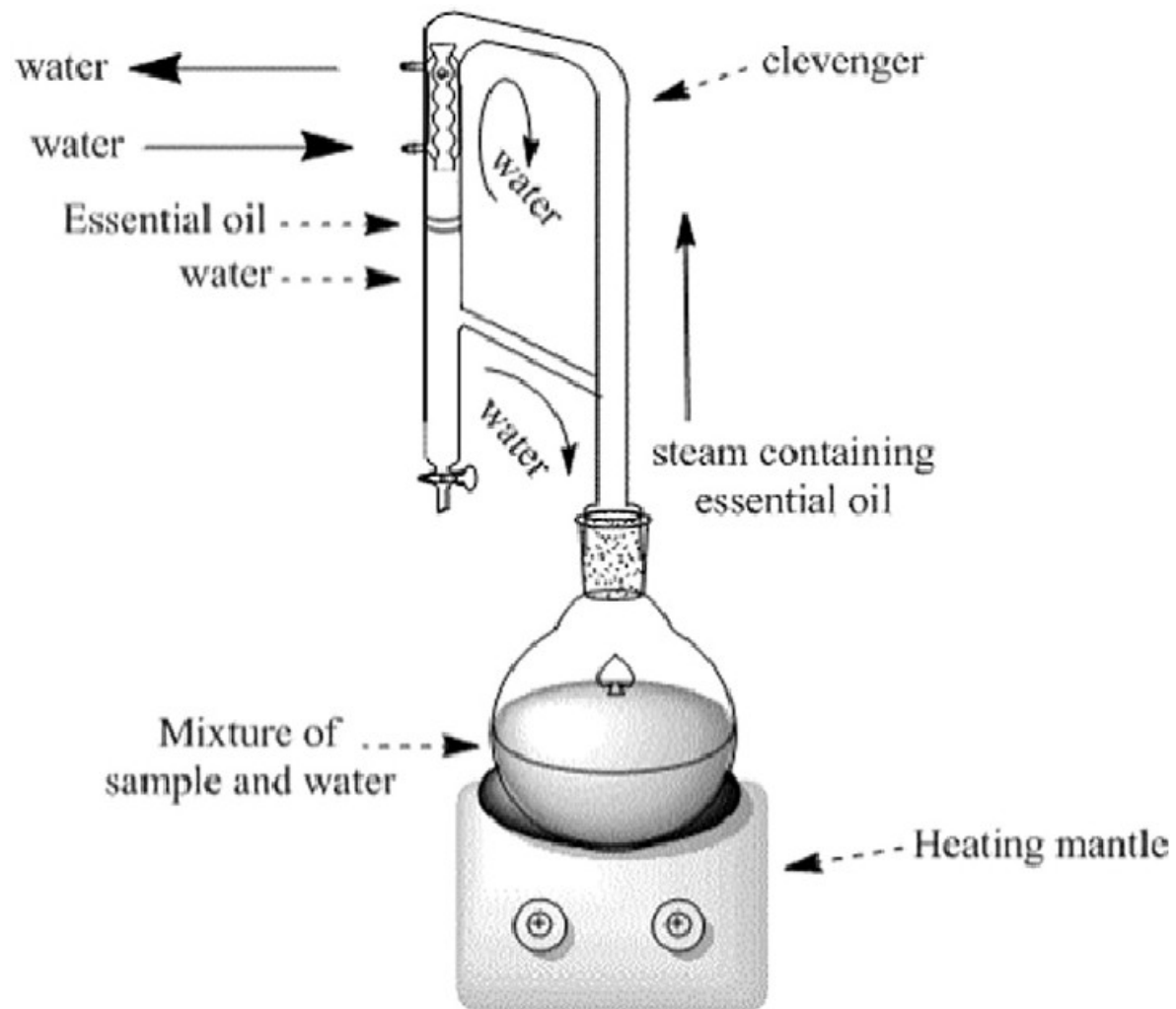
SUPERCRITICAL FLUIDS....

- Ideal characteristics of supercritical fluids
 - is a natural product
 - Chemically inert
 - Non-flammable
 - Perfectly non –toxic,
 - Easy to eliminate completely
 - Selective, Readily available and Inexpensive

DISTILLATION

- **i. Water distillation**
- This method is applied to plant material that is dried and not subject to injury by boiling e.g.: turpentine oil is obtained in this manner.
- Crude turpentine oleoresin is introduced with water into the distilling chamber and boiled with water until both volatile matter and water, are condensed.

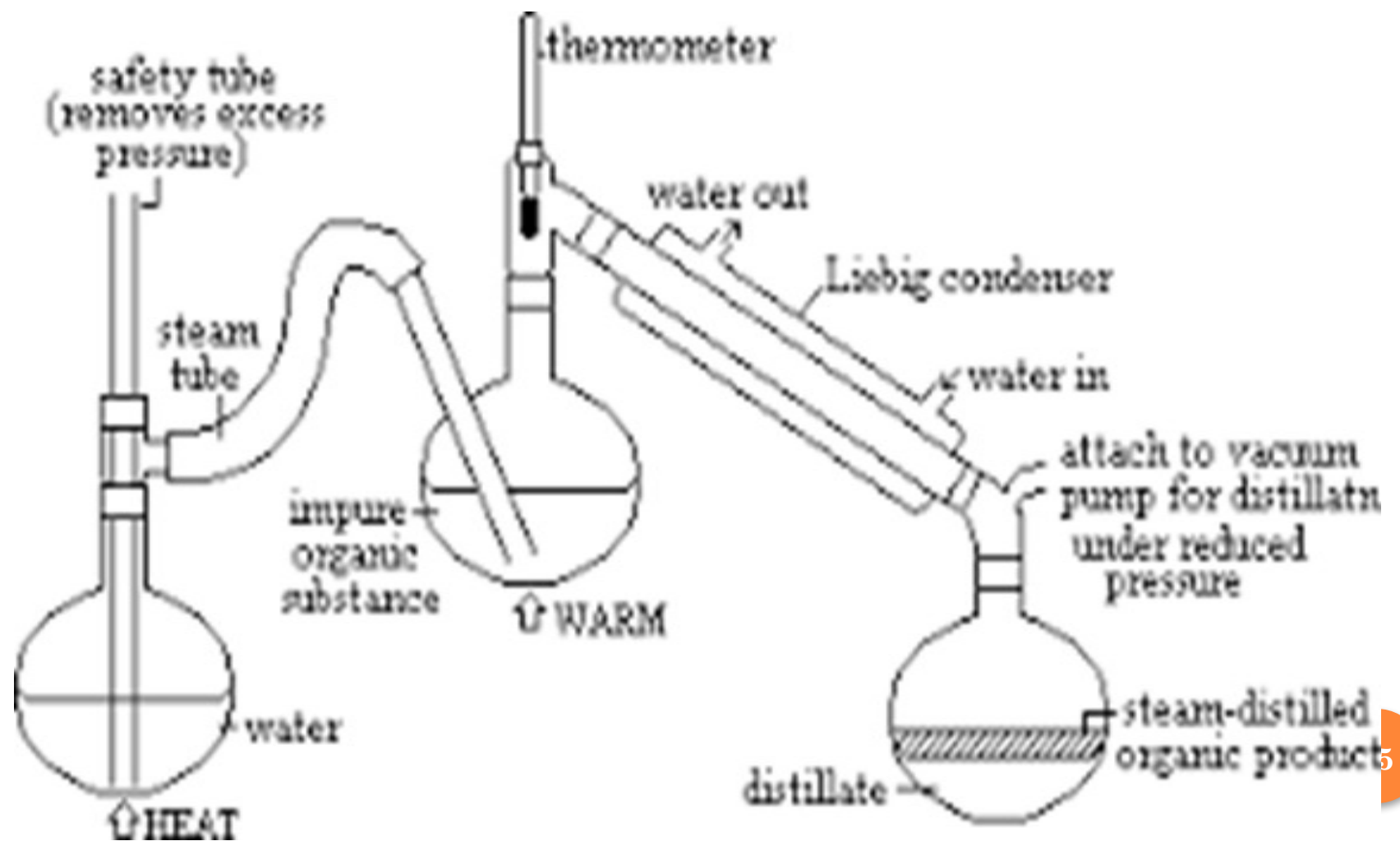
HYDRODISTILLATION

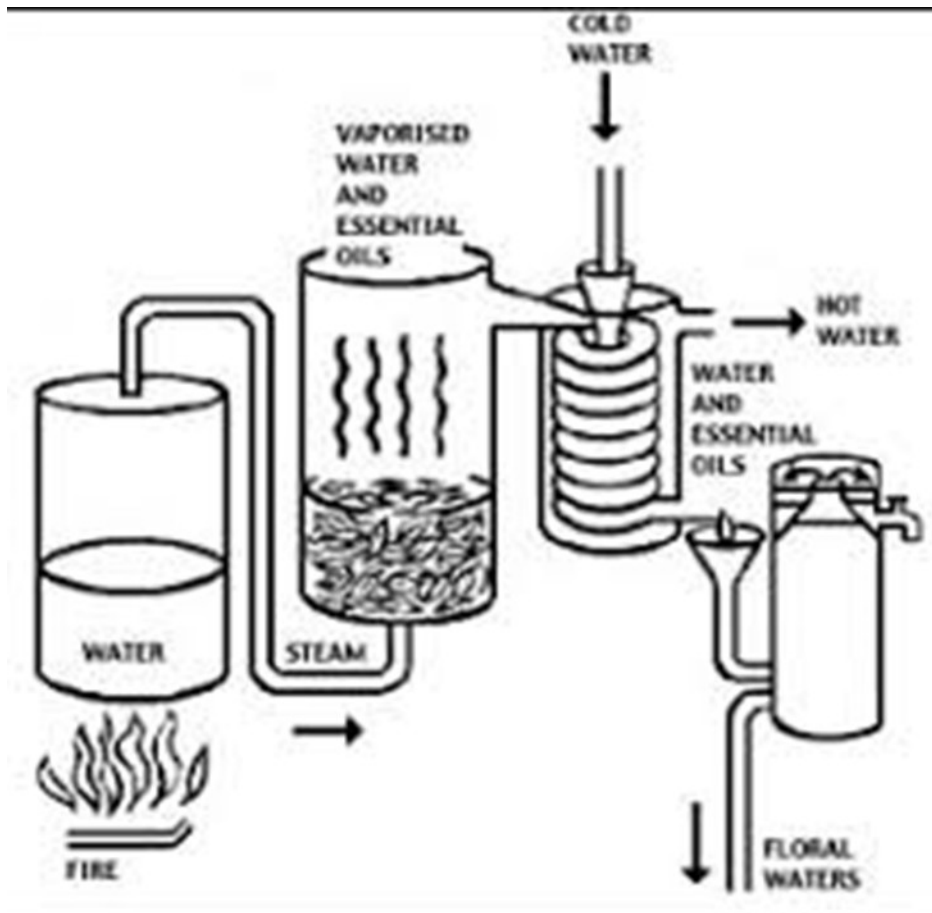


DISTILLATION...

○ II. Direct steam distillation

- This method is applicable for fresh drugs e.g.: peppermint.
- Crop is cut and taken directly to the distilling chambers.
- Drug is then suspended in wire basket or perforated trays or a similar container (grid).
- Steam is forced through the herb carrying the oil droplets to the condensing chamber.





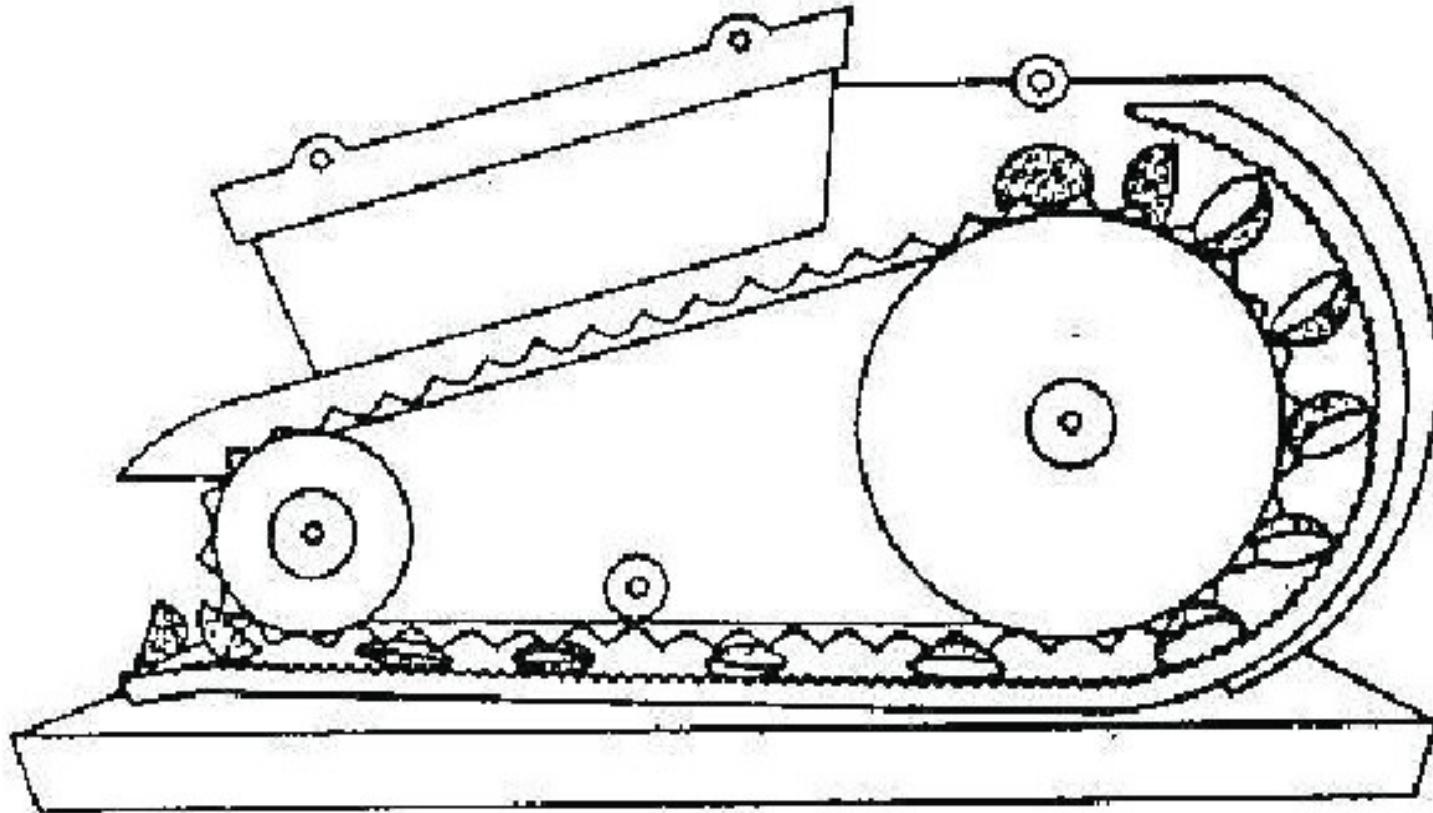
MECHANICAL EXTRACTION/EXPRESSION

- The extrusion by application of pressure (squeezing out).
- Volatile oils obtained by the expression methods maintain their fragrance while those obtained by distillation are inferior.
- Expression of volatile oils can be carried out by means of any of the following methods
 - The Sponge method
 - The Scarification method [Ecuelle à piquer]

EXPRESSION...

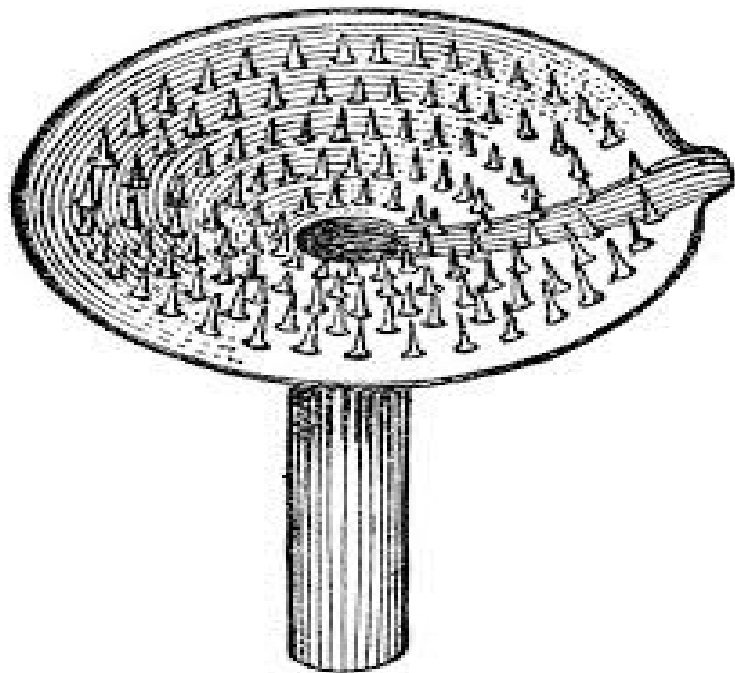
- **a. The sponge method:**
- The citrus fruit cut into halves and the pulp is removed.
- The rind is turned out and squeezed and the oil collected by a sponge
- The sponge is then periodically squeezed in vessel.
- The upper oily layer in the vessel is separated.

THE SPONGE METHOD



EXPRESSION...

- **b- The écuelle à piquer (ecuelle a piquer) process:**
- The entire fruits are rotated in a saucer-shaped dish provided with metal pins or sharp metal projections
- The exuding oil and water are collected in depression in the bottom of the saucer.
- The liquid is poured off into a large vessel, where it is allowed to stand until the oil can be decanted and separated.



○ **THANK YOU**